

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

06th November 2015

Subject ST8 - General Insurance: Pricing

Time allowed: Three hours (14.45* – 18.00 Hrs)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

- 1) *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception*
- 2) ** You have 15 minutes at the start of the examination in which you are required to read the questions. You are strongly encouraged to use this time for reading only, but notes may be made. You then have three hours to complete the paper.*
- 3) *You must not start writing your answers in the answer sheet until instructed to do so by the supervisor.*
- 4) *The answers are not expected to be any country or jurisdiction specific. However, if examples/illustrations are required for any answer, the country or jurisdiction from which they are drawn should be mentioned.*
- 5) *Attempt all questions, beginning your answer to each question on a separate sheet.*
- 6) *Mark allocations are shown in brackets.*
- 7) *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** You work in the Pricing department of a large General Insurance company which is a market leader in personal lines of business. A senior colleague of yours has mentioned that “offset” is a very helpful and versatile option in the context of Generalised Linear Modelling.

Suggest two scenarios in modelling where “offset” can be used. For each of these, outline how you would like to use “offset” in modelling.

[4]

- Q. 2)** In the wake of increased regulatory supervision and consumer awareness, corporates from different industries have started enquiring about Product Recall insurance.

i) List down the possible insured events which are likely to give rise to a product recall.

(2)

ii) Outline the types of coverage under Product Recall Insurance.

(3)

iii) Identify two key features of product recall insurance which differentiate it from product liability insurance.

(2)

[7]

- Q. 3)** The reinsurance structure of an insurer writing commercial property insurance is described below.

The insurer has a quota share treaty for the entire portfolio. The quota share treaty features sliding scale commissions in which the commission paid by the reinsurer depends on the actual loss ratio of the underlying business.

The risk excess of loss treaties are also purchased. One treaty is for a working layer and the other one is for a higher layer. These risk XoL treaties apply after the quota share treaty.

There is also catastrophe aggregate excess of loss reinsurance treaty in place, which applies after quota share and risk XoL treaties, in that order. The occurrence retention is 5 million and occurrence limit is 20 million. This treaty has no aggregate retention and also no aggregate limit.

i) Briefly describe the possible reasons why sliding scale commission might be preferred to profit commission by the reinsurer for quota share reinsurance.

(3)

ii) Comment on suitability of using the sliding scale commission or profit commission arrangement to the risk excess of loss treaties purchased.

(2)

The flood losses under the catastrophe reinsurance has “hours clause” which specifies the limited period of 72 hours during which claims can be aggregated for the purpose of one claim on the reinsurance contract. The insurer is allowed to choose the starting point of the period to which the reinsurance claim can apply.

Suppose that due to floods ranging over the course of four days, the portfolio suffered losses. Assume that the floods hit at the exactly same time on all four days. Assume that day 1 witnessed 10 million in claims, day 2 witnessed 8 million in claims, day 3 witnessed 15 million and day 4 witnessed 12 million in claims. All of these claim amounts are after the application of quota share followed by the Risk XoL treaties

- iii)** Identify how the insurer should define the event/events in order to maximize the recovery from the catastrophe reinsurance. (2)
[7]

- Q. 4) i)** An insurer is choosing between two risk classification options for a specific cover. One of these risk classification options is more detailed and results in a price for each risk that is closer to the actual cost, when compared with the other classification option. Briefly explain the main considerations to be made by the insurer in choosing one risk classification option between the two. (3)

- ii)** A large insurer sells a variety of both personal and commercial lines of insurance. It is trying to identify third-party data that is not specific to insurance and use it to improve the risk classification for pricing. List any five examples of such types of data sources and, for each example, briefly state how such data could be used to improve the risk classification. (5)
[8]

- Q. 5)** You are pricing a risk excess of loss reinsurance treaty for employer's liability business of an insurer. For the experience rating approach, the insurer provided the historical premiums, claims and also the historical rate changes along with loss trends for that portfolio.

- i)** The rate change indices were produced by aggregating the observed rate changes on individual renewals. Briefly describe the considerations to be made in applying the on-level premium factors base on these rate changes to bring the historical premiums to the current rate levels. (3)
- ii)** Briefly describe how the reinsurer could treat the shock losses in the cedant's loss history for the purpose of pricing the risk excess of loss treaty. (2)

The cedant provided the loss history using the same reporting threshold, which is half of the retention limit, for all the claims in the history.

- iii)** Outline the advantages in instead obtaining the history of all claims whose trended amounts, trended to bring them to the current cost levels, exceed the threshold. (2)

The reinsurer is using a certain proportion of the standard deviation of the loss as the risk load.

- iv)** Outline the possible limitation of this approach. (1)
[8]

- Q. 6)** You are provided the following empirical summary data of all 2000 number of claims by the insurer toward bodily injury liability cover of motor insurance business for your company. Assume that no deductible or excess is applicable to the claims from this cover. Also assume that only a per-claim limit is applicable. In the table below, summary is provided for all the historical claims, separately based on the per-claim limit applied.

Claim Size Band	Per-Claim Limit	Number of Claims	Amount of Claims
0-99999	100000	810	35,855,808
0-99999	250000	394	17,800,556
0-99999	500000	208	8,732,804
100000	100000	326	32,600,000
100001-249999	250000	135	20,276,866
100001-249999	500000	68	10,138,725
250000	250000	42	10,500,000
250001-499999	500000	14	4,550,252
500000	500000	3	1,500,000

- i) Calculate the average claim severity for per-claim limits of 100000, 250000 and 500000. Identify and state the limitation in the deriving the ILF based on the average claim severities. (3)
- ii) By estimating the expected severity for the layers 0 to 100000, 100000 to 250000 and 250000 to 500000, derive the ILFs for per-claim limits of 250000 and 500000 with 100000 as the base per-claim limit. (4)

A new rule mandates that the victims suffering severe bodily injury be provided some benefit for “suffering” in addition to the medical and nursing costs.

- iii) Comment on how this is expected to impact the ILFs. (2)
[9]

- Q. 7)** You are pricing a large commercial policy as a blend of premiums as per experience rating and exposure rating methods.

- i) Outline the factors that influence the choice of the weighting between the pure experience rated premium and the exposure rated premium. (5)
- ii) State two arguments supporting the use of experience rating in general insurance. (2)
- iii) State two factors, other than price, that are also important to a large business, in comparison to an individual, when choosing its insurer. (2)
[9]

- Q. 8)** The number of claims from the j^{th} risk ($j = 1, 2, \dots, n$) is either 0 or 1 and the probability of a claim arising from is 'p' for all risks. Losses from these risks are independent. Claim amounts, in the event of a loss, from any risk, follow exponential distribution with parameter $1/\theta$, with PDF $\frac{1}{\theta} e^{-\frac{x}{\theta}}, x > 0$.

- i)** Determine the mean and variance of the aggregate claims distribution. (2)

Suppose the actual values of 'p' and ' θ ' are not known but their estimates are known. It is also known that the actual values could lie anywhere, independent of each other, with uniform probability, within +/- 10% of their estimated values which are $\hat{p}$ and $\hat{\theta}$ respectively.

- ii)** Comment on why the new variance of the aggregate claims distribution under this scenario would be different from part (i). (2)

- iii)** Write down the steps involved in using the simulation approach to find the estimates for the mean and standard deviation of the aggregate claims distribution under this scenario. (5)

[9]

- Q. 9)** You work for the Underwriting and Product Pricing function in a general insurance company which is considering rate changes on its motor book of business. The Pricing Director wants to estimate the impact that a rate change will have on its policyholder retention and the resulting profitability of the transformed book.

- i)** In order to develop a model around this, what type of information you would consider as model input? (2)
- ii)** For a specific set of 250 policyholders having comprehensive motor insurance cover from your Company, you have the following information:

Current premium	INR 500
Premium of Competitors	
Competitor 1	INR 610
Competitor 2	INR 452
Competitor 3	INR 564
Competitor 4	INR 498

You also have got the following two tables about how existing customers are likely to shop around and switch to other insurer.

Premium increase	Chance to shop around
INR 0 and below	3%
INR 1 to 50	5%
INR 51 to 100	25%
INR 101 to 200	70%
INR 201 and above	100%

Premium savings	Chance to switch
INR 0	8%
INR 1 to 50	15%
INR 51 to 100	40%
INR 101 to 200	96%
INR 201 and above	100%

The Company is considering increasing the premium either by 9% or 12%. Assuming total losses and expense per policy from this set as INR 472, find out which rate increase will be preferable for your Company.

(4)

- iii) The Pricing Director is also planning to build a Profit Optimisation model for new business. You are given a Benefit function which is defined as

Benefit function = Conversion rate x (Gross premium – Risk premium – Expenses)

Using the above function, describe briefly how you would develop such a model.

(3)

[9]

- Q. 10)** A medium sized General Insurer writes mainly motor and health business. Understanding the huge potential in the under-penetrated home insurance sector, the Company Management has decided to focus towards developing adequate pricing strategy for this line.

While doing that, their Chief Underwriter is very nervous about the impact of fluctuating inflation level and possible catastrophic events on home insurance business.

- i) Discuss the different types of claims inflation which can impact the home insurance claims cost. (3)
- ii) You work for the Catastrophe Modelling team within the same Organisation. Referring to the basic structure of a catastrophe model, describe how you would model earthquake as a key peril to this line of business. (5)
- iii) You are also asked to prepare a write-up on the usage of catastrophe models. Identify different functional areas in insurance operations where catastrophe models are used. For each of these areas, write down how such a model can be utilised. (6)

[14]

- Q. 11)** A large General Insurance company writing multiple lines of business, have experienced steady decrease in conversion rate in its personal motor insurance business. Over last two years, their comprehensive book has shrunk, directly impacting the top line growth of the company.

Preliminary investigations at a high level have indicated towards out-of-line premiums being charged from the customers who purchased insurance through internet. Management has decided to establish a process of competitor pricing analysis on a regular basis in order to gain insights into the premiums charged by the competitors in the market.

As the pricing actuary, you have been assigned the task of establishing a framework for competitor price deconstruction exercise. You have to figure out the different rating factors and their corresponding levels used by the competitors and also to determine the rate relativities. You know that currently a large number for rating factors are used to price comprehensive motor insurance business. Also there is no regulatory restriction in companies trying to deconstruct competitor's prices.

- i) List down the possible sources of writing motor insurance business. (1)
- ii) Define Conversion Rate. (1)
- iii) In order to work on such an assignment, what type of data you would require? Suggest a practical way of collecting the required information and also comment on any potential challenges you may encounter while sourcing the data. (5)
- iv) Set out the approach of carrying out a competitor rate deconstruction exercise, putting more emphasis around three primary rating factors – driver, vehicle and location. You can keep age fixed and find out the relativities for other factors and associated levels. (6)
- v) Management has also got a feeling that, the Company probably misread the stage of insurance cycle during the stated period. Defining the term “insurance cycle”, suggest possible causes of this phenomenon in the present context. (3)

[16]
